Composition: One Function Inside Another

f(4)

Composition: $(f \circ g)(x) = f(g(x))$. Apply the inside function first.

$$f(x) = x^2 + 1, \quad g(x) = 2x - 3.$$

Find $(f \circ g)(x)$, $(g \circ f)(x)$, and $(f \circ g)(2)$. Are the two compositions equal?

- $(f \circ g)(x)$ is just a new notation. plug $g(x)$ into $f(x)$.
- $(f \circ g)(x) = f(g(x))$

$= (\ \cancel{g(x)}\ )^2 + 1$

$2x-3$

$= (2x-3)^2 + 1 = 4x^2 - 12x + 9 + 1 = 4x^2 - 12x + 10$

- $(g \circ f)(x) = g(f(x)) = 2(x^2+2) - 3$

$= 2x^2 + 2 - 3 = 2x^2 - 1$

- $(f \circ g)(2) = 4(2)^2 - 12 \cdot (2) + 10$

$= 16 - 24 + 10 = 2$

Composition Domains: Keep Inner Restrictions

An input must be in the domain of the **inside** function, and its output must be allowed in the **outside** function.

$$f(x) = \frac{1}{x}, \quad g(x) = \frac{3x}{x^2 - 1}.$$

Find $(f \circ g)(x)$ and its domain. Do not lose restrictions when simplifying.

$$f(g(x)) = \frac{1}{\frac{3x}{x^2-1}} = \frac{x^2-1}{3x}$$

domain $x \neq 2, -1, 0$

$x \xrightarrow{g} g(x) \xrightarrow{f} f(g(x))$

$x \longrightarrow f(x) = y$

- domain of $f(g(x))$ must satisfied the domain of $g(x)$.

  Domain for g is $x \neq 2, -1$

- $f(x) = \frac{1}{x}$, 0 is not allowed.

  $\frac{3x}{x^2-1} \neq 0$

  $\frac{3x}{3} \neq \frac{0}{3}$

  $x \neq 0$

Composition with Square Roots

Let $f(x) = \sqrt{x-2}$ and $g(x) = x^2 - 1$. Find both compositions and the domain of each.

$$(f \circ g)(x) \qquad (g \circ f)(x).$$

A real square root requires a nonnegative radicand.

- $f(g(x)) = \sqrt{(x^2-1)-2} = \sqrt{x^2-3}$

$x^2 - 1 \geq 0$

$x^2 \geq 1$

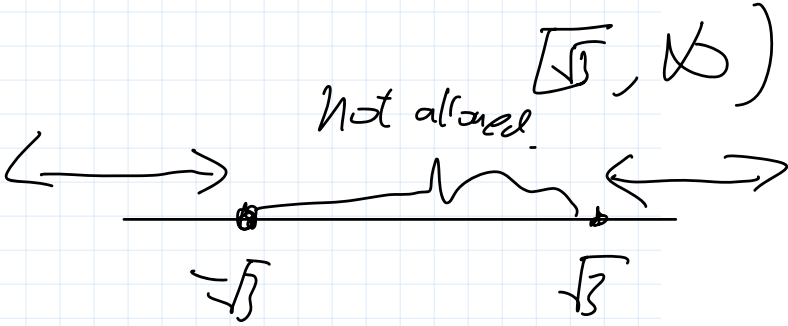
$x \leq -1$ or $x \geq 1$

$x^2 - 3 \geq 0$

$x^2 \geq 3$

$x \leq -\sqrt{3}$ or $x \geq \sqrt{3}$

Domain of $f(g(x))$: $(-\infty, -\sqrt{3}] \cup [\sqrt{3}, \infty)$



- $(g \circ f)(x) = g(f(x)) = (\sqrt{x-2})^2 - 1$

$= x - 2 - 1 = x - 3$.

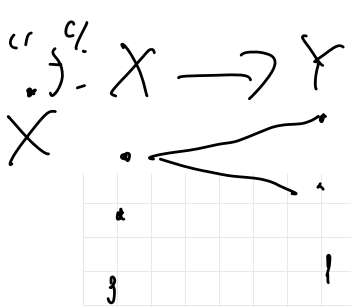
$x - 2 \geq 0$

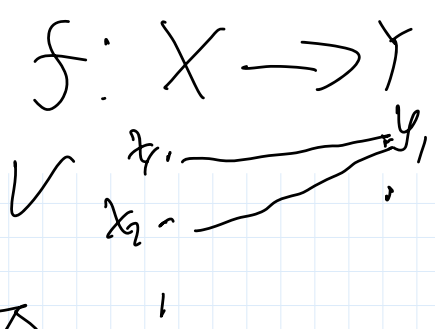
$x \geq 2$

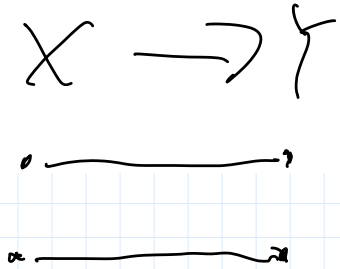
$[2, \infty)$

One-to-One Functions

A function is **one-to-one** if different inputs have different outputs. Its graph passes the horizontal line test. Decide whether $f(x) = x^2$ and $g(x) = x^3$ are one-to-one on $\mathbb{R}$.



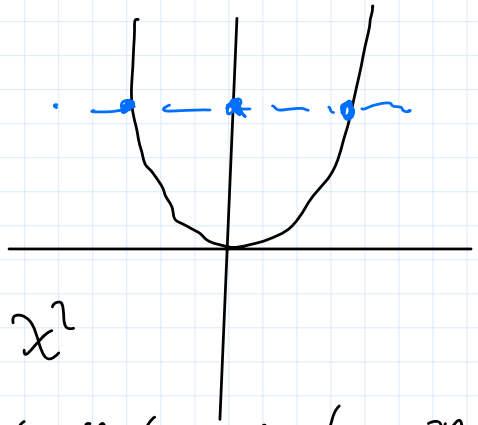




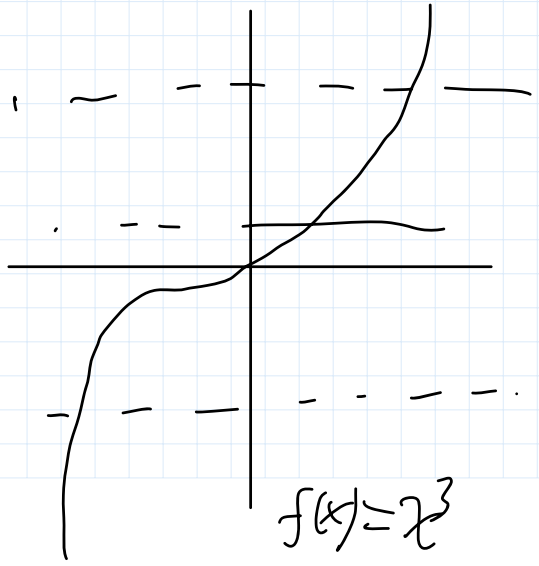
$f(x) = x^2$

$f(-1) = 1$

$f(1) = 1$



is not one-to-one



one-to-one function.

Find and Check an Inverse

Find f^{-1} for $f(x) = 2x + 5$, then check both compositions. For a second example, find g^{-1} for $g(x) = (x-2)^3$. **Method:** write $y = f(x)$, interchange x and y, then solve for y.

- $f(x) = 2x + 5 \xrightarrow{\text{inverse}} x = 2y + 5$

$y = 2x + 5$

$x = 2y + 5$

$\frac{x-5}{2} = \frac{2y}{2}$

$f^{-1}(x) = \frac{x-5}{2}$

$g(x) = (x-2)^3 \Rightarrow x = (y-2)^3$

$y = (x-2)^3$

$\sqrt[3]{x} = y - 2$

$\sqrt[3]{x} + 2 = f^{-1}(x)$

Restrict a Domain Before Inverting

The full quadratic $x^2 + 2$ is not one-to-one. Use the restricted function

$$f(x) = x^2 + 2, \qquad x \geq 0.$$

Find its inverse. State both domains and ranges, and check both compositions on their correct domains.

- $y = x^2 + 2 \Rightarrow$ inverse $\Rightarrow x = y^2 + 2$

$$\sqrt{x-2} = \sqrt{y^2}$$

$$f^{-1}(x) = y = \sqrt{x-2}$$

	Domain	Range
$f(x) = x^2 + 2$	$[0, \infty)$	$[2, \infty)$
$f^{-1}(x) = \sqrt{x-2}$	$[2, \infty)$	$[0, \infty)$

$$f(f^{-1}(x)) = (\sqrt{x-2})^2 + 2 = x - 2 + 2 = x$$

$$f^{-1}(x) = \sqrt{x-2}$$

$$f^{-1}(f(x)) = \sqrt{f(x) - 2} = \sqrt{x^2 + 2 - 2} = \sqrt{x^2} = x$$

Inverse Graphs Reflect Across a Line

Sketch $f(x) = x^2 + 2$ for $x \geq 0$, its inverse $f^{-1}(x) = \sqrt{x-2}$, and the line $y = x$. Use the points $(0, 2)$, $(1, 3)$, and $(2, 6)$ on f to locate three points on its inverse.

$f(x) = x^2 + 2$	$f^{-1}(x) = \sqrt{x-2}$
$(0, 2)$	$(2, 0)$
$(1, 3)$	$(3, 1)$
$(2, 6)$	$(6, 2)$

